

Unimodality of the Fourth-Order q -Fibonomial Coefficients

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Abstract

Let $F_0 = 0$, $F_1 = 1$, and $F_{r+1} = F_r + F_{r-1}$. We prove that $\begin{bmatrix} m+4 \\ 4 \end{bmatrix}_{F,q}$ has a symmetric weakly unimodal coefficient sequence for every integer $m \geq 1$. Thus the fixed fourth-order slice of the q -Fibonomial unimodality conjecture holds for all parameters. The proof reduces every lower-half first coefficient difference to four values of the $\{1, 2, 3\}$ -part denominator. A six-residue formula gives uniform positivity for $m \geq 7$, and the remaining six cases are verified by exact integer coefficient recomposition. No assertion is made about the unrestricted two-parameter conjecture.

1 Context and statement

For a positive integer N , set

$$[N]_q = 1 + q + \cdots + q^{N-1}, \quad [F_N]!_q = \prod_{j=1}^N [F_j]_q,$$

and define

$$\begin{bmatrix} m+n \\ n \end{bmatrix}_{F,q} = \frac{[F_{m+n}]!_q}{[F_m]!_q [F_n]!_q}.$$

Bergeron, Ceballos, and Küstner gave a weighted-tiling interpretation of these polynomials and formulated their unimodality conjecture [1]. Connelly, Ito, Martinez, Shevchenko, and Yang proved the cases $n \leq 3$ and reported exact computations in a finite range [2]. The result below treats the next fixed slice. The comparison is deliberately limited to those inspected sources; it is not a priority claim beyond them.

Theorem 1. *For every integer $m \geq 1$, write*

$$P_m(q) = \begin{bmatrix} m+4 \\ 4 \end{bmatrix}_{F,q} = \sum_{k=0}^{D_m} c_{m,k} q^k.$$

Then

$$c_{m,k+1} - c_{m,k} \geq 0 \quad \text{for} \quad 0 \leq k < \left\lfloor \frac{D_m}{2} \right\rfloor.$$

Moreover, $c_{m,k} = c_{m,D_m-k}$; hence the coefficient sequence is symmetric and weakly unimodal.

The unrestricted conjecture, in which both m and n vary, is not addressed here. Our argument is algebraic and does not provide a chain decomposition or another combinatorial explanation of unimodality.

2 A denominator

For $n \geq 0$, define

$$h(n) = [q^n] \frac{1}{(1-q)(1-q^2)(1-q^3)} = \#\{(x, y, z) \in \mathbb{Z}_{\geq 0}^3 : x + 2y + 3z = n\},$$

and put $h(n) = 0$ for $n < 0$.

Lemma 2. *For every $n \geq 0$,*

$$h(n) = \left\lfloor \frac{n^2 + 6n + 12}{12} \right\rfloor = Q(n) - \epsilon(n), \quad Q(n) = \frac{n^2 + 6n + 12}{12},$$

where $0 \leq \epsilon(n) \leq 7/12$. More precisely, for $n \equiv 0, 1, 2, 3, 4, 5 \pmod{6}$, respectively,

$$\epsilon(n) = 0, \frac{7}{12}, \frac{1}{3}, \frac{1}{4}, \frac{1}{3}, \frac{7}{12}.$$

The function h is nondecreasing on $\mathbb{Z}_{\geq 0}$.

Proof. Summing first over z gives

$$h(n) = \sum_{z=0}^{\lfloor n/3 \rfloor} \left(\left\lfloor \frac{n-3z}{2} \right\rfloor + 1 \right).$$

Writing $n = 6u + s$ and evaluating this finite sum yields

s	$h(6u + s)$	$Q(6u + s) - h(6u + s)$
0	$3u^2 + 3u + 1$	0
1	$3u^2 + 4u + 1$	$7/12$
2	$3u^2 + 5u + 2$	$1/3$
3	$3u^2 + 6u + 3$	$1/4$
4	$3u^2 + 7u + 4$	$1/3$
5	$3u^2 + 8u + 5$	$7/12$

This proves the formula and the error bounds. Finally, $(x, y, z) \mapsto (x+1, y, z)$ injects the solutions counted by $h(n)$ into those counted by $h(n+1)$. $\square$

3 Lower-half coefficient differences

Fix m and put

$$a = F_{m+1}, \quad b = F_{m+2}.$$

Thus $F_{m+3} = a + b$ and $F_{m+4} = a + 2b$. Cancelling the factorials in the definition gives

$$P_m(q) = \frac{[a]_q [b]_q [a+b]_q [a+2b]_q}{[2]_q [3]_q} = \frac{(1-q^a)(1-q^b)(1-q^{a+b})(1-q^{a+2b})}{(1-q)^2(1-q^2)(1-q^3)}. \quad (1)$$

For completeness, this quotient is a polynomial in this fixed slice. The Fibonacci recurrence modulo 2 and modulo 3 shows that F_r is even when $3 \mid r$ and is divisible by 3 when $4 \mid r$. Among

four consecutive indices there is an index of each type. Therefore the numerator product of q -integers in (1) is divisible by both $[2]_q$ and $[3]_q$. These two polynomials are coprime, so their product divides the numerator. Since every q -integer is reciprocal, (1) also gives

$$q^D P_m(q^{-1}) = P_m(q), \quad D = 3a + 4b - 7. \quad (2)$$

Let $M = \lfloor D/2 \rfloor$ and suppress the first subscript by writing $P_m(q) = \sum c_k q^k$. The numerator in (1) expands exactly as

$$\begin{aligned} & (1 - q^a)(1 - q^b)(1 - q^{a+b})(1 - q^{a+2b}) \\ &= 1 - q^a - q^b + q^{2a+b} + q^{a+3b} - q^{3a+3b} - q^{2a+4b} + q^{3a+4b}. \end{aligned} \quad (3)$$

Because

$$2(a + 3b) - D = 2b - a + 7 > 0,$$

every term of (3) beginning at degree $a + 3b$ lies above M . Taking coefficients of $(1 - q)P_m(q)$ therefore gives, for every $1 \leq t \leq M$,

$$c_t - c_{t-1} = g(t) := h(t) - h(t - a) - h(t - b) + h(t - L), \quad L = 2a + b. \quad (4)$$

4 Uniform positivity

Suppose first that $m \geq 7$. Then $a \geq F_8 = 21$ and $b \geq a + 1$. We verify (4) in four exhaustive ranges.

If $t < a$, then $g(t) = h(t) \geq 0$. If $a \leq t < b$, monotonicity in Lemma 2 gives $g(t) = h(t) - h(t - a) \geq 0$.

If $b \leq t < L$, set

$$R_3(t) = Q(t) - Q(t - a) - Q(t - b).$$

Lemma 2 gives $g(t) \geq R_3(t) - 7/12$. The quadratic R_3 is concave, so its minimum on the integer interval $b \leq t \leq L - 1$ occurs at an endpoint. Direct substitution yields

$$12R_3(b) = a(2b - a + 6) - 12, \quad 12R_3(L - 1) = a(2b - a - 4) - 7.$$

The first numerator exceeds the second by $10a - 5$. Hence

$$g(t) \geq \frac{a(2b - a - 4) - 14}{12} \geq \frac{a(a - 2) - 14}{12} > 0. \quad (5)$$

Finally, if $L \leq t \leq M$, all four arguments in (4) are nonnegative. With

$$R_4(t) = Q(t) - Q(t - a) - Q(t - b) + Q(t - L),$$

the signs of the four error terms in Lemma 2 give the uniform loss $7/6$, and expansion gives

$$g(t) \geq R_4(t) - \frac{7}{6}, \quad R_4(t) = \frac{a(3a + 4b - 2t - 6)}{12}.$$

Since $2t \leq D = 3a + 4b - 7$,

$$g(t) \geq \frac{a - 14}{12} > 0. \quad (6)$$

The intervals above may be empty, but together they cover every $1 \leq t \leq M$. Thus all required differences are nonnegative for $m \geq 7$.

5 The six base cases

It remains to check $1 \leq m \leq 6$. This is a finite exact-integer calculation. To make the check reproducible, form

$$A_m(q) = \prod_{r=m+1}^{m+4} [F_r]_q = \sum_j A_j q^j$$

by ordinary coefficient convolution. Since $[2]_q[3]_q = 1 + 2q + 2q^2 + q^3$, low-to-high division is the integer recurrence

$$c_k = A_k - 2c_{k-1} - 2c_{k-2} - c_{k-3}, \quad c_j = 0 \quad (j < 0). \quad (7)$$

For each row below, exact recomposition checked $A_j = c_j + 2c_{j-1} + 2c_{j-2} + c_{j-3}$ at every degree, and all coefficients outside $0 \leq j \leq D$ were set to zero. The resulting lower-half differences have the following minima.

m	(a, b)	D	number checked	$\min_{1 \leq t \leq M} (c_t - c_{t-1})$
1	(1, 2)	4	2	0
2	(2, 3)	11	5	0
3	(3, 5)	22	11	0
4	(5, 8)	40	20	0
5	(8, 13)	69	34	1
6	(13, 21)	116	58	1

All arithmetic in this check is integer arithmetic; no numerical approximation enters. Exact coefficient generation followed by analytic estimates is also used in a separate enumerative-probability setting in [3]. That citation records only this adjacent methodological relationship: it contains no result about Fibonomials and is not used in the proof above.

Combining the base cases with (5)–(6) proves $c_t - c_{t-1} \geq 0$ for every $1 \leq t \leq M$ and every $m \geq 1$. Reciprocity (2) then proves symmetry and weak unimodality, completing the proof of Theorem 1.

Computational and AI assistance disclosure

Proof exploration and manuscript preparation used OpenAI Codex language-model assistance. The six finite cases were computed with Python arbitrary-precision integers, with complete coefficient recomposition, and were independently reconstructed during a separate mathematical audit. The uniform argument is symbolic; the finite computation is used only for $m = 1, \dots, 6$. The author reviewed the stated argument, scope, citations, and disclosure and is responsible for the manuscript.

References

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